

Impact of Electrical and Mechanical Antenna Tilt on LTE Downlink System Performance

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Abstract—Antenna tilt is one of the most important performance tuning parameters of a cellular network, since it has a strong impact on the inter-site interference level in the system. In this paper, we present an analysis of the impact of antenna tilt on LTE coverage and capacity. Using system simulations, we study how the distribution between two types of tilt, electrical and mechanical, affects path gain and cell edge, peak, and average throughput in a macro-cellular scenario. While the total tilt has a strong impact on both capacity and coverage, we find that the type of tilt has distinct impact only on capacity.

I. INTRODUCTION

Base station antenna tilting is a common technique for improving cell isolation and/or increasing coverage in cellular networks [1]–[4]. Tilt is an important design parameter when considering coverage vs. capacity during cell planning as well as when tuning live networks. It can be used together with, and independently of, other interference reduction techniques such as inter-cell interference coordination (ICIC) [5].

Tilt can be achieved electrically, mechanically, or by a combination thereof [6]. Remote tilt, which allows non-disruptive tuning of live networks, is typically implemented using RET (remote electrical tilt) antennas. Due to grating lobe effects, RET antenna tilt intervals are typically limited to $<10^\circ$ relative a nominal tilt direction, which may be insufficient in cell plans with dense site positions and/or high antenna installations. A total tilt larger than this can then be achieved by applying mechanical tilt to a RET antenna, to get a tilt interval tailored for a given scenario.

Mechanical tilt means that the antenna is physically rotated around an axis, typically horizontal, which changes the *effective* radiation pattern (as viewed from ground) but leaves the radiation pattern *per se* unchanged. Electrical tilt is achieved by applying a phase (or time) taper to the element excitations, which introduces changes both in the effective radiation pattern and in the radiation pattern *per se*. Since effective radiation pattern behavior depends on tilt type, differences with respect to system performance may occur. Analysis of system performance impact of joint electrical and mechanical tilt is therefore of great interest.

The impact of electrical *or* mechanical tilt on system performance has been investigated for GSM [1] and WCDMA [2]–[4]. In [3] electrical tilt was shown to be a key factor for improving downlink performance in WCDMA, while [4] identified differences regarding the impact of tilt type on

system performance. In LTE, with a frequency reuse factor of one, no intra-cell interference, and no macro diversity, tilt is likely to be even more important for achieving good cell isolation and, hence, high system performance.

Recently, Yilmaz *et al.* presented an analysis of the impact of joint electrical and mechanical tilt on LTE system performance [7]. They found that electrical tilt gives higher capacity than mechanical tilt and that tilt type has impact on optimal tilt angle. The present paper extends this work by:

- finding optimal combinations of electrical and mechanical tilt for a wide range of azimuth and elevation beamwidths;
- presenting a sensitivity analysis that shows the performance loss if pure electrical or pure mechanical tilt is used instead of the optimal combination;
- presenting a simple model of system performance, which is validated against a detailed dynamic system simulator;
- validating the 3GPP antenna model against measured patterns for a wide range of tilt combinations;
- using the updated, accurate, 3GPP mechanical tilt model.

II. SYSTEM MODEL

The focus of this paper is on relative system performance in the downlink for different tilt settings, not on performance predictions in absolute numbers. This means that a fairly simple model of system performance can be used, since all details that do not effect relative performance can be ignored.

A. System Performance Model

In this study, both the base station, or evolved node B (eNB), and the user equipment (UE) have a single antenna even though LTE will employ multi-antenna techniques. When all individual antennas in a multi-antenna configuration share the same radiation pattern characteristics, such as beamwidths and sidelobe levels, the assumption is that the relative impact of tilt on system performance is similar for single- and multi-antenna configurations. We have found support for this conjecture by comparing single- and multi-antenna configurations in more detailed dynamic system simulations.

The system performance model is based on computation of the downlink signal-to-interference-plus-noise ratio (SINR) distribution in a target cell, i.e., for all users served by a specific base station antenna, in the presence of a number of non-target cells served by other antennas. We assume that the transmitted downlink power per physical resource

block (PRB) is the same for all PRBs and all UEs in the network, and also that the network is fully loaded such that this power is transmitted in all PRBs in all cells in the network. We further assume that UEs are allocated full system bandwidth in a Round Robin fashion and that the network is deployed with a frequency reuse factor of one. Assuming a frequency-independent radio channel, we can analyze system performance by calculating the SINR per user for a single PRB, since all PRBs for a user will have the same SINR. The SINR for UE n (in any PRB) is thus simply calculated as

$$\text{SINR}_n = \frac{P g_{1,n}}{P \sum_{c=2}^M g_{c,n} + N_0}, \quad (1)$$

where P is the transmitted downlink power per PRB and $g_{c,n}$ is the path gain from the eNB antenna in cell c to UE n and, specifically, $g_{1,n}$ denotes the line-of-sight path gain from the base station antenna serving the target cell (cell number 1) to UE n in said cell. *Path gain* is here defined as antenna gain divided by path loss including lognormal fading. Cell selection is based on strongest path gain, regardless of actual user position. Finally, M is the number of cells in the simulated network, and N_0 is the thermal noise power per PRB.

Path gain is a position-dependent measure of relative signal strength. In the coverage analysis, the system is assumed to be noise-limited, i.e., $P \sum_{c=2}^M g_{c,n} \ll N_0$. Since N_0 is constant, we choose to define *coverage* simply as the 5-percentile target cell path gain, and coverage can then be considered a measure of cell edge signal strength performance.

In the capacity analysis, the system is assumed to be interference limited, i.e., $P \sum_{c=2}^M g_{c,n} \gg N_0$. Motivated by Shannon's capacity formula, we approximate the spectral efficiency for UE n , C_n (bps/Hz), by

$$C_n = \log_2(1 + \text{SINR}_n). \quad (2)$$

Since we are only interested in relative performance, we choose this spectral efficiency as a measure of throughput.

The target cell coverage and throughput distributions are obtained by sampling a surface containing multiple eNB sites uniformly over a regular grid and computing the coverage and throughput measures for each sample point belonging to the target cell, which is done for multiple lognormal fading realizations. The computed performance measures can then be used to compute a CDF over the target cell, or more concentrated measures such as averages or CDF percentiles.

B. Antenna Model

The base station antenna radiation pattern is modeled in two cardinal cuts; an azimuth pattern with relative gain $G_{\text{az}}(\phi)$ (dB) and an elevation pattern with relative gain $G_{\text{el}}(\alpha)$ (dB). These 1-D patterns are modeled by a Gaussian-shaped main beam with a sidelobe floor according to

$$G_{\text{az}}(\phi) = \max \left(-12 \left(\frac{\phi}{\text{HPBW}_{\text{az}}} \right)^2, \text{SLL}_{\text{az}} \right), \quad (3)$$

$$G_{\text{el}}(\alpha) = \max \left(-12 \left(\frac{\alpha + \alpha_e}{\text{HPBW}_{\text{el}}} \right)^2, \text{SLL}_{\text{el}} \right), \quad (4)$$

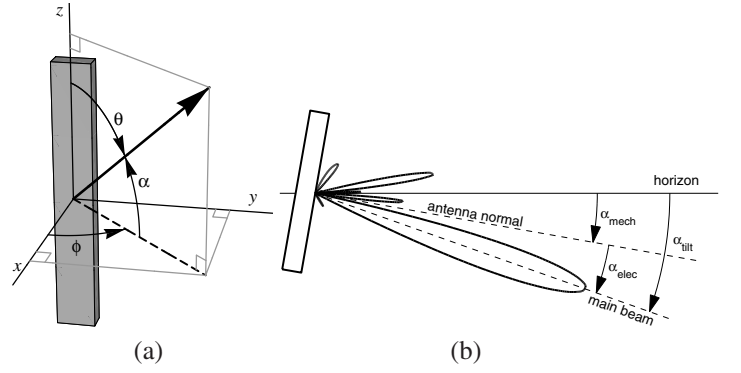


Fig. 1. Angles definitions: (a) spherical angles θ and ϕ , and elevation angle α , for a given direction from a base station antenna; (b) electrical tilt α_e , mechanical tilt α_m , and total tilt α_{tilt} angles for an antenna tilted in the vertical plane containing the main beam peak.

where $\phi, -\pi \leq \phi \leq \pi$, is the azimuth angle and $\alpha, -\pi/2 \leq \alpha \leq \pi/2$, is the elevation angle related to the polar angle θ as $\alpha = \pi/2 - \theta$ in an antenna-fixed coordinate system with its z -axis parallel to the antenna cylinder axis, see Fig. 1(a). Furthermore, α_e is the electrical downtilt (positive when tilting below the xy -plane, i.e., the horizontal plane for a vertical z -axis), and HPBW and SLL (< 0 ; dB) are the half-power beamwidth and sidelobe level for the respective patterns. The antenna gain in an arbitrary direction (α, ϕ) is modeled as

$$G(\alpha, \phi) = \max \{ G_{\text{az}}(\phi) + G_{\text{el}}(\alpha), \text{SLL}_0 \} + G_0, \quad (5)$$

for an overall sidelobe floor SLL_0 (dB) and peak antenna gain G_0 (dBi). For the interval of electrical tilt values considered here, the impact on the radiation pattern directivity will be negligible, and we therefore use a constant value for G_0 . This antenna model has also been proposed by 3GPP to be used in system simulations [8].

Mechanical tilt is modeled using the updated 3GPP model [8] which represents a coordinate transformation between spherical coordinates (θ', ϕ') in an Earth-fixed coordinate system and the antenna-fixed coordinates (α, ϕ) defined by

$$\begin{aligned} \alpha &= \pi/2 - \arccos(\cos \phi' \sin \theta' \sin \alpha_m + \cos \theta' \cos \alpha_m), \\ \phi &= \arg(\cos \phi' \sin \theta' \cos \alpha_m - \cos \theta' \sin \alpha_m + j \sin \phi' \sin \theta'), \end{aligned}$$

where α_m is the mechanical tilt angle. In contrast to the previous 3GPP mechanical tilt model [9], the updated tilt model preserves the radiation pattern shape, obeys conservation of energy, and supports polarized fields (not used in this study).

Finally, the total tilt α_{tilt} in the vertical plane containing the beam peak, and orthogonal to a horizontal axis of rotation, is the sum of the electrical and mechanical tilts as illustrated in Fig. 1(b). We let r be the ratio of electrical to total tilt:

$$r = \alpha_e / \alpha_{\text{tilt}} = \alpha_e / (\alpha_e + \alpha_m). \quad (6)$$

We note that pure electrical tilt produces an elevation steering of the radiation pattern which is independent of horizontal direction (azimuth angle in an Earth-fixed coordinate system) whereas mechanical tilt does not. Hence, the horizontal half-power beamwidth, and thus the relative radiated power density

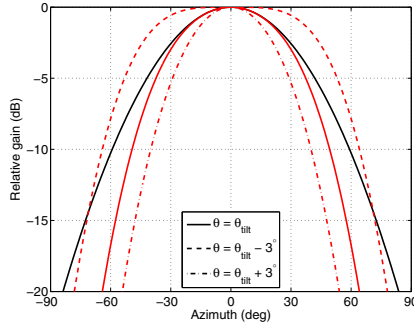


Fig. 2. Antenna gain on ground as a function of azimuth angle, for different conical cuts around a vertical axis with each curve peak-normalized. Gain for electrical tilt is independent of elevation angle, while mechanical tilt affects the beamwidth and shape differently for each elevation angle.

TABLE I
DEFAULT PARAMETER SETTINGS USED IN THE SIMULATIONS.

Base station height	30 m
Mobile height	1.5 m
Intersite distance	500 m
#sectors/site	3
#sites in network	19
Path loss	$134 + 35 \log_{10} R$ dB, R in km
Lognormal fading standard deviation	8 dB
Fading correlation between different sites	0.5
Percentage indoor users	0%
Antenna gain, G_0	18 dBi
Elevation HPBW, $HPBW_{el}$	6.5°
Elevation SLL, SLL_{el}	-17 dB
Azimuth HPBW, $HPBW_{az}$	65°
Azimuth SLL, SLL_{az}	-25 dB
SLL floor, SLL_0	-30 dB
eNB power per PRB, P	29 dBm
Noise power per PRB, N_0	-111 dBm

(on downlink), depends on the vertical angle for mechanical tilt, as shown in Fig. 2, while the beamwidth is constant for electrical tilt. This suggests that mechanical and electrical tilt may have different impact on system performance.

III. PERFORMANCE ANALYSIS

A. Simulation Setup

A number of cells surrounding the target cell is used in order to generate an interference environment. The simulated network consists of 19 3-sector macro sites placed on a hexagonal grid and with the sector antennas pointing to the neighbor site. We assume that all eNBs in the network have identical antennas and tilt settings. Table I summarizes the parameter settings that have been used in the simulations.

B. Coverage

Coverage (5-percentile path gain) calculated for all combinations of electrical tilt $\alpha_e \in [-5, 15]^\circ$ and mechanical tilt $\alpha_m \in [-5, 20]^\circ$ is plotted in Fig. 3(a), normalized to the peak coverage value, with reference traces for three different

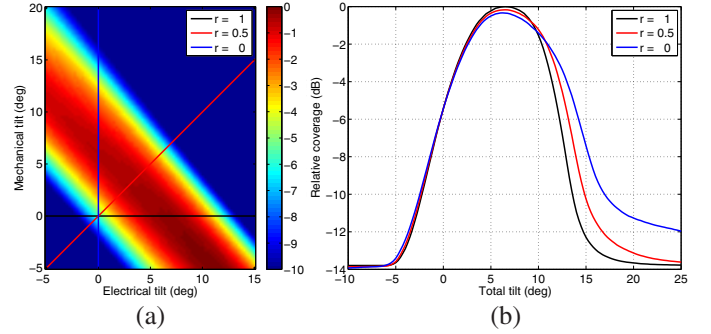


Fig. 3. Relative coverage plotted against (a) electrical and mechanical tilt and (b) total tilt for three different tilt type combinations.

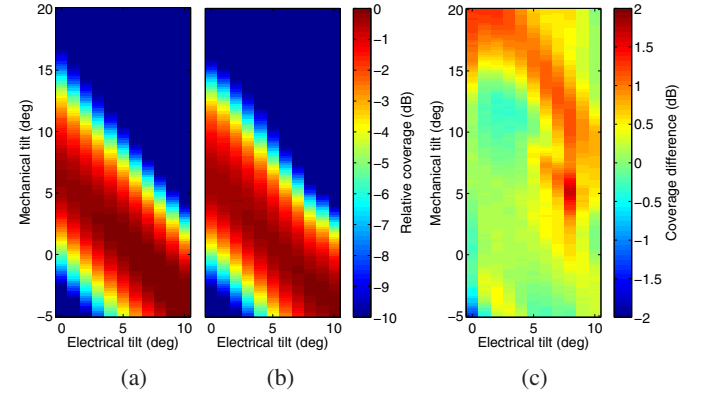


Fig. 4. Relative coverage for (a) radiation pattern model and (b) full-sphere measurement data for Kathrein 742215, and (c) difference between model pattern and measured pattern coverage.

tilt type combinations shown: pure electrical ($r = 1$), pure mechanical ($r = 0$), and equal amounts of electrical and mechanical tilt ($r = 0.5$). Fig. 3(b) shows the path gain as a function of total tilt along the three reference traces. These graphs show that the total tilt setting has a large impact on coverage, while the tilt type combination has little impact on optimal coverage (less than 0.5 dB).

The coverage results for the simple radiation pattern model in (3)–(5) are validated against results for measured radiation patterns of a common sector antenna, the Kathrein antenna 742215 [10]. The coverage was calculated using full-sphere measurement data from 1700 MHz to 2200 MHz and the results were averaged over frequency and antenna port (polarization) for the available electrical tilt values of $\{0, 1, \dots, 10\}^\circ$. Fig. 4 shows the coverage for the radiation pattern model and measured pattern data. The agreement is good, with about 1 dB or less difference in coverage for all tilt combinations. This indicates that the pattern model is sufficiently detailed, and with relevant parameter settings, a valid representation of real antenna behavior for coverage calculations.

Although coverage is defined as the 5-percentile path gain, it is also interesting to consider the path gain behavior for other percentiles. The optimized tilt for each percentile is shown in Fig. 5 for the three electrical tilt ratios, $r = 0, 0.5$, and 1. The conclusion is that tilt type combination has only negligible impact on optimized tilt with respect to path gain.

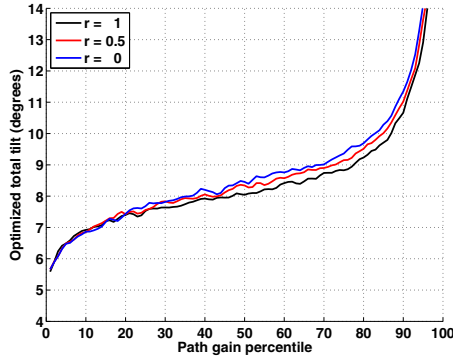


Fig. 5. Optimized total tilt vs. path gain percentile for three different tilt type combinations: $r = 1$ (electrical tilt only), $r = 0.5$ (equal amounts of electrical and mechanical tilt), and $r = 0$ (mechanical tilt only).

C. Capacity

The metrics used in the capacity evaluation are:

- 5-percentile of the throughput CDF. This can be used as a measure of cell-edge bit rate;
- mean of all throughput values in the cell. This can be used as a measure of cell throughput;
- 95-percentile of the throughput CDF. This can be used as a measure of peak bit rate.

Since we are only interested in relative performance, we normalize all throughput values to the maximum value for each considered parameter sweep.

The system performance model described in Section II is a simple one. Yet, we have found it to be a powerful tool for rapid evaluation of relative system performance. To give some credibility to this analysis, Fig. 6 shows a comparison of results from this simple model with results from a detailed dynamic system simulator which includes models of, for example, scheduling, adaptive coding and modulation, UE mobility, and delays in channel quality reports. It also contains an implementation of the 3GPP spatial channel model (SCM) [11]. The results show relative throughput vs. mechanical and electrical tilt. Clearly, the simple system model gives similar predictions of relative system performance as the dynamic system simulator.

Fig. 7 shows how the different throughput metrics depend on the total tilt for the three different electrical tilt ratios, $r = 0, 0.5$, and 1 . Clearly, the total tilt has a strong impact on all considered capacity metrics. Regarding optimal tilt type combination, the results show that for cell edge (5%) and mean throughput pure electrical tilt is optimal, while pure mechanical tilt gives lowest performance. The results also show that the antennas should be tilted less with electrical tilt than with mechanical. For peak rate (95%), an equal amount of electrical and mechanical tilt is optimal. In this case, pure electrical tilt has the lowest performance. The antennas should be tilted less with electrical tilt than with mechanical also for peak rate. Another observation is that cell edge performance is more sensitive to tilt than peak rate. It is also interesting to note that the optimal total tilt for cell edge bit rate is one half

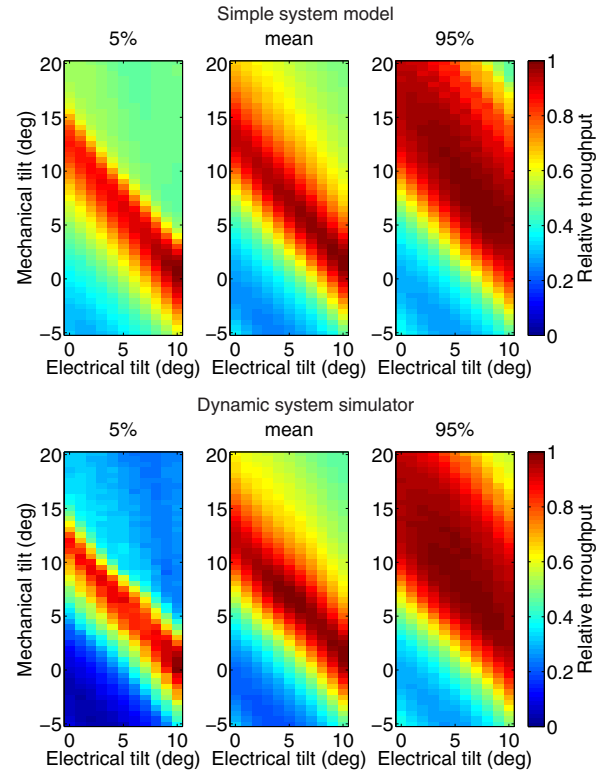


Fig. 6. Relative throughput vs. electrical and mechanical tilt.

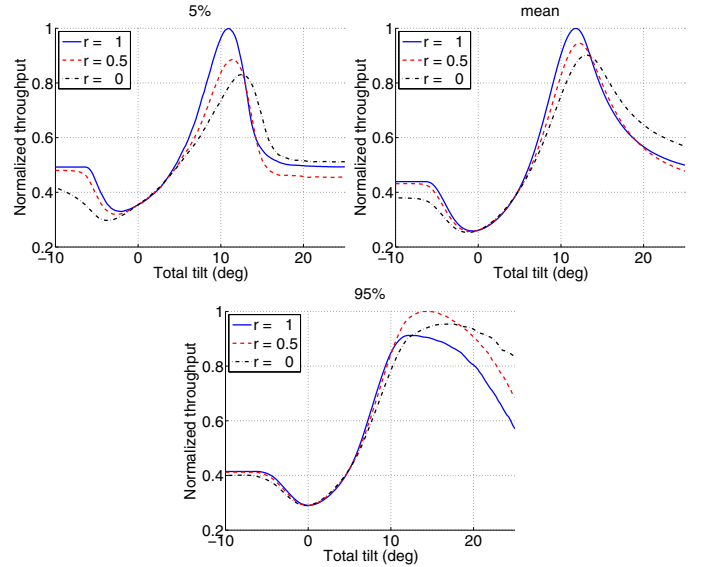


Fig. 7. Normalized throughput vs. total tilt for different tilt combinations.

HPBW less than optimal tilt for peak rate.

The optimal tilt combination may depend on other antenna parameters such as the beamwidths of the azimuth and elevation patterns. To illustrate the robustness of the previous conclusions to such variations, Fig. 8 shows how the optimal electrical tilt ratio, r , depends on the azimuth and elevation HPBWs for the different performance metrics. When the azimuth HPBW is varied, the elevation HPBW is fixed at its

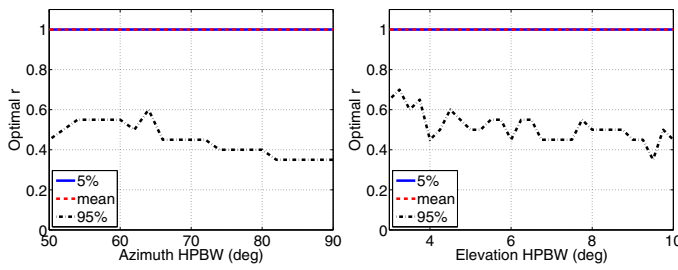


Fig. 8. Optimal tilt combination and relative throughput loss vs. azimuth beamwidth.

default value, and vice versa. In the considered scenario, the system is interference limited, thus the antenna gain can be kept constant while beamwidths are changed. For cell edge and mean throughput the optimal electrical tilt ratio is 1 (one), i.e., pure electrical tilt, for all HPBWs. For peak throughput the optimal tilt ratio is in the range 0.4-0.6, i.e., roughly equal amounts of electrical and mechanical tilt for all HPBWs.

Another robustness issue to consider is how sensitive performance is to a correct combination of electrical and mechanical tilt. Fig. 9 shows the loss in throughput if pure mechanical or pure electrical tilt is employed relative to the throughput obtained when they are combined optimally. For each value of the HPBW the throughput for the optimal combination for this HPBW is normalized to 100%. Since electrical tilt is optimal for cell edge and mean throughput for all HPBWs, the loss for electrical tilt is 0% in these cases. With mechanical tilt the loss compared to the optimal tilt combination, i.e. pure electrical tilt, is up to 25% for cell edge and up to 10% for mean throughput. For peak throughput the loss is up to 25% for pure electrical tilt and up to 7% for pure mechanical tilt. A general observation is that cell edge performance is the most sensitive performance metric with regard to choice of tilt type.

IV. CONCLUSION

In this paper we have shown how LTE downlink system performance is affected by different combinations of electrical and mechanical tilt of the eNB antenna. The analysis has been carried out using model radiation patterns and a simple model of system performance. These have been validated against measured patterns and a dynamic system simulator. With respect to coverage, the conclusion is that the choice of tilt method, or combination of tilt methods, has insignificant impact, and that the optimal total (electrical + mechanical) tilt is similarly insensitive to choice of tilt method.

For capacity, a careful division of the total tilt into electrical and mechanical is more important. Pure electrical tilt is optimal for cell edge and mean throughput, while equal amounts of electrical and mechanical tilt is optimal for peak rate. This conclusion holds for a wide range of elevation and azimuth beamwidths. The differences in optimal throughput between different combinations of tilt methods is at most 25%, cell edge performance being the most sensitive to tilt type combination. The results also confirm the previously known results that total tilt has strong impact on both coverage and capacity.

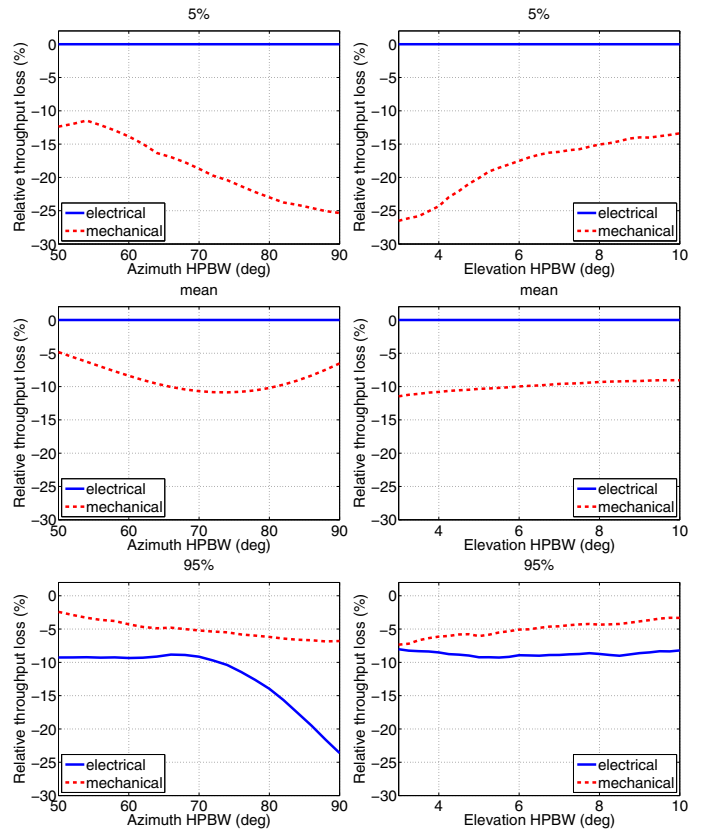


Fig. 9. Optimal tilt combination and relative throughput loss vs. azimuth beamwidth.

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